

Simplifying Algebraic Fractions (Homework)

Simplify.

Answers

$$\textcircled{1} \frac{x-2}{x^2+2x-8}$$

$$\frac{1}{x+4}$$

$$\textcircled{2} \frac{y^2+y-20}{y+5}$$

$$y-4$$

$$\textcircled{3} \frac{b^2+8b+15}{b^2+6b+9}$$

$$\frac{b+5}{b+3}$$

$$\textcircled{4} \frac{x^2-7x-8}{x^2-5x-24}$$

$$\frac{x+1}{x+3}$$

$$\textcircled{5} \frac{2y^2-23y+45}{y^2-81}$$

$$\frac{2y-5}{y+9}$$

$$\textcircled{6} \frac{4b^2-1}{6b^2+5b+1}$$

$$\frac{2b-1}{3b+1}$$

$$\textcircled{7} \frac{x+1}{x^2-1}$$

$$\frac{1}{x-1}$$

$$\textcircled{8} \frac{8y^2+59y+21}{y+7}$$

$$8y+3$$

★ Use
the -1
rule.

$$\textcircled{9} \frac{10-b}{b^2-13b+30}$$

$$\frac{-1}{b-3}$$

$$\textcircled{10} \frac{x^2+6x-27}{3-x}$$

$$\frac{-(x+9)}{\text{or } -x-9}$$

$$\textcircled{11} \frac{b^3+5b^2+6b}{b^4-3b^3-18b^2}$$

$$\frac{b+2}{b(b-6)}$$

$$\textcircled{12} \frac{5y^3-20y}{20y^2-40y}$$

$$\frac{y+2}{4}$$

$$\textcircled{13} \frac{-4b^5+28b^4-24b^3}{12-12b^2}$$

$$\frac{b^3(b-6)}{3(b+1)}$$

$$(14) \frac{2x^4 - 4x^3 + 4x^2 - 8x}{2x^6 + 4x^4}$$

$$\frac{x-2}{x^3}$$

$$(15) \frac{-2b^7 + 12b^5 - 10b^3}{12b^4 + 32b^3 + 20b^2}$$

$$-\frac{b(b-1)(b^2-5)}{2(3b+5)}$$

$$(16) \frac{32y^5 - 18y^7}{3y^5 - y^4 - 4y^3}$$

$$\frac{-2y^2(3y+4)}{y+1}$$

$$(17) \frac{y^4 - y^2 - 12}{5y^4 + 10y^3 + 7y + 14}$$

$$\frac{(y-2)(y^2+3)}{5y^3+7}$$

$$(18) \frac{2a^3b^2 - 6a^2b^3 - 36ab^4}{6a^3b + 12a^2b^2 - 18ab^3}$$

$$\frac{b(a-6b)}{3(a-b)}$$

$$(19) \frac{64 - x^2}{x+8}$$

$$\begin{array}{l} -(x-8) \\ \text{or } -x+8 \\ \text{or } 8-x \end{array}$$

$$(20) \frac{3+y}{y^2+7y+12}$$

$$\frac{1}{y+4}$$

Hint: $y+3=3+y$

$$(21) \frac{-3x^4 + 9x^3 + 30x^2}{5x^3 - x^4}$$

$$\frac{3(x+2)}{x}$$

$$(22) \frac{3x^2y - 243y^3}{81y^3 - 9xy^2}$$

$$-\frac{x+9y}{3y}$$

$$(23) \frac{b^4 - 8b^2 + 16}{b^2 - 4b + 4}$$

$$(b+2)(b+2)$$

$$(24) \frac{25 - x^2}{5 - x}$$

$$\begin{array}{l} 5+x \\ \text{or } x+5 \end{array}$$

$$(25) \frac{y^2 + 12y + 11}{1 - y^2}$$

$$\begin{array}{l} \frac{y+11}{-(y-1)} \\ \text{or} \\ \frac{y+11}{-y+1} \\ \text{or} \\ \frac{y+11}{1-y} \end{array}$$